



STARDOG **DESIGNER**

Designer Training



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Course Overview

Description:

- This beginner training is for users that are responsible for designing, building and maintaining data models. The training will discuss the basics of data modeling and will teach users how to connect resources through mapping. Users will learn how to build out a model, publish it and then view it in Explorer. They will learn how to build out inference rules and then apply these rules to their data model.

Audience:

- New Stardog users that are consumers, modelers or builders of knowledge graphs.

Pre-requisites:

- User should have a Stardog Free Cloud account and endpoint
- Users should complete the Meet Stardog and Explorer training

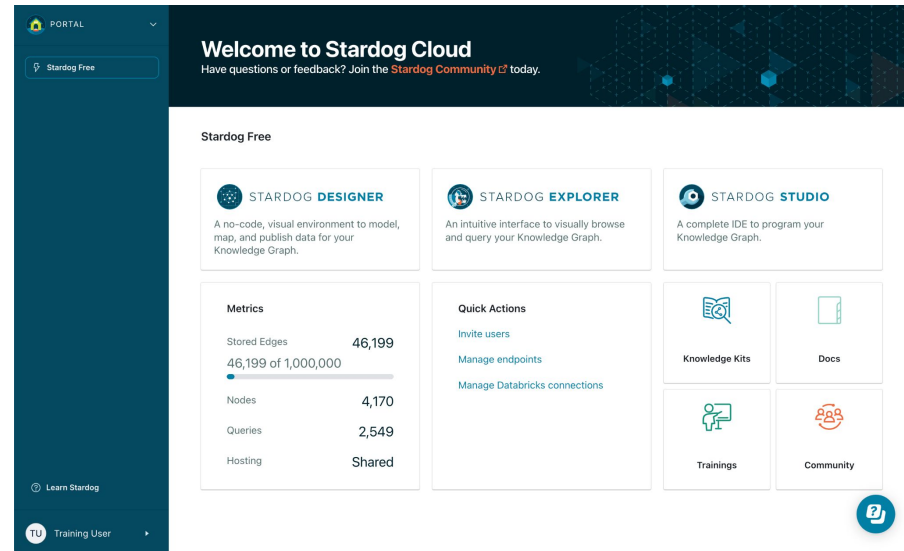
Course Objectives

After completing this training, students will be able to:

- Understand basic data modeling and terminology
- Create mappings to project resources
- Create and map relationships
- Create and edit attributes and labels
- Publish a model
- Create and publish inference rules

Stardog Cloud

- Manage Knowledge Graph endpoints
- Launch applications
- Sign up for Stardog Cloud plans
- Access Resource Center for guided journeys and training resources
- Invite more users



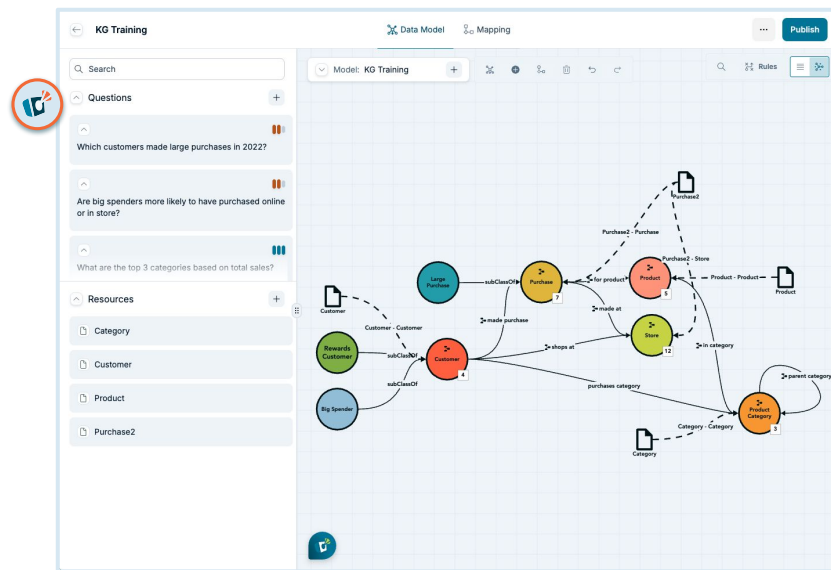
Stardog Designer

Stardog Designer

No-code, visual KG authoring

Features:

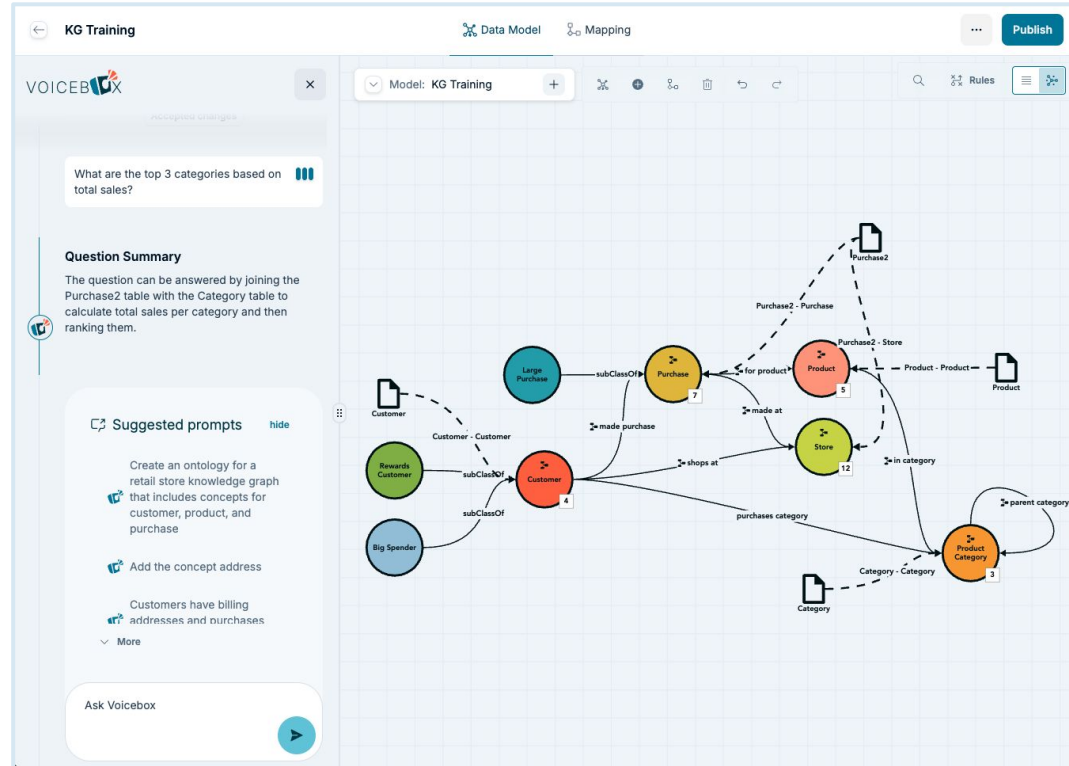
- Create & Edit data models
- Load existing models
- Define inference rules
- Knowledge Catalog powered mapping



Voicebox in Designer

Use Voicebox as an expert user that can help you:

- Create new concepts based on core concepts
- Extend and modify the data model
- Map project resources to the data model
- Available for Voicebox enabled endpoints
- Voicebox only features highlighted with 



Agenda

Data Modeling

Project Resources & Mapping

Relationships

Publish the Model

Inference Rules



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Top Down Example



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Use Case Overview

The use case for this course is based on a **Customer 360 analysis**.

The customer in this case is someone that buys products from the company's ecommerce site or in a brick and mortar store.

Here are some questions that users will ask of this use case:

- What do customers buy and how often?
- Do customers use loyalty accounts?
- What type of payment methods do they use?
- What are the buying patterns for customers based on location, age, gender, etc?
- What products are typically bought together?
- Are there any patterns related to when the customer buys in a store vs. online?



Identify Business Concepts

In Designer, the user will create the data model to support the use case. Before using Designer, think about the data model and start to identify business concepts.

Customers

Purchase

Loyalty Account

Payments

Store Location

Gender

Product

Year

Customer Location

Store Type



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Create Ontological Statements

Use the business concepts to form ontological statements. For example, Customer buys products

- Customer made purchase
- Customer has a Loyalty Account
- Products are in a category
- Categories have a parent category
- Customer shops in a physical store
- Customer shops online
- Customer is female
- Purchase made with a visa
- Purchase made in 2022
- In store shoppers are customers who have an in store purchase



Noun & Verb Analysis

Identify the **nouns** and **verbs**:

- Customer **made** a purchase
- Customer **made** purchase with a Loyalty Account
- Products **are** in a category
- Categories **have** a parent category
- Customer **shops** in a physical store
- Customer **shops** online
- Customer **is** female
- Purchase **made** with a visa
- Purchase **made** in 2022
- *In store shoppers* are **Customers** who have an *in store purchase*

In Designer the nouns will turn into Classes, Attributes or literal values and the verbs will become relationships.



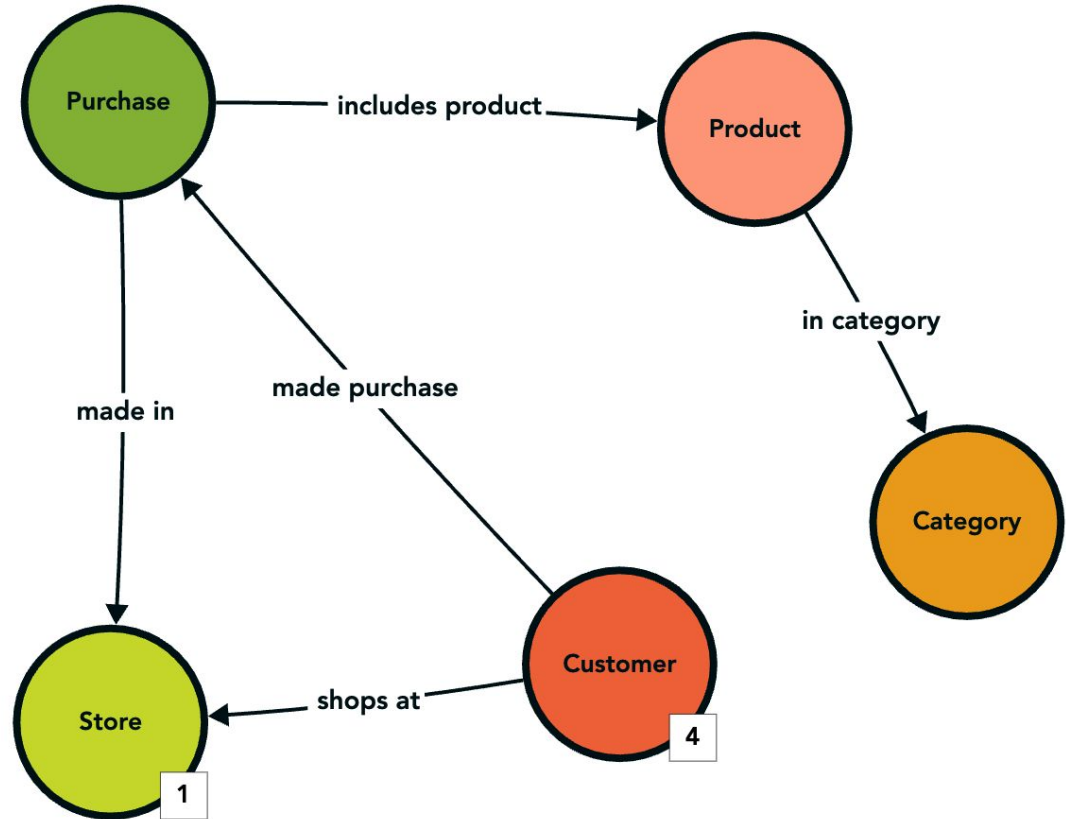
Identify the Attributes

Identify the noun (subj), verbs and attributes:

- Customer **made** a purchase
- Customer **made** purchase with a Loyalty Account
- Products **are** in a category
- Categories **have** a parent category
- Customer **shops** in a Store [physical store]
- Customer **shops** at a Store [online]
- Customer **is** Gender [female]
- Purchase **made** with a Credit Card[visa, mastercard, amex]
- Purchase **made** in Year [2022]
- *In store shoppers* are Customers who have an *in store purchase*

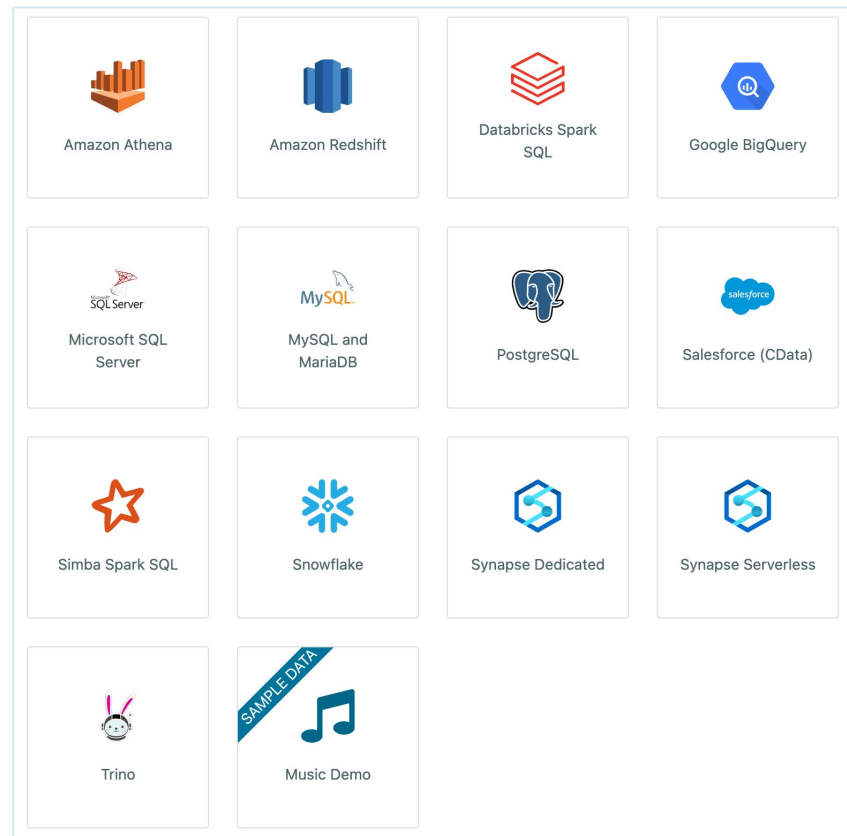
Sketch the Data Model

- Create Classes for the nouns
- Connect the Classes with the verbs (relationships)
- Create a list of attributes for each Class



Identify Data Sources

- Think about what data is needed to populate these concepts
- Create a list of these sources
- Determine if this data should be virtualized or materialized or some combination of the two



Data Modeling Overview



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Getting Started with Your First Data Model

- Start with a few key concepts or your data
- Data models should be composable like Legos, reuse is key
- Naming
 - Data models encode the meaning of your data
 - Terminology should align with domain experts



Data Modeling Approaches

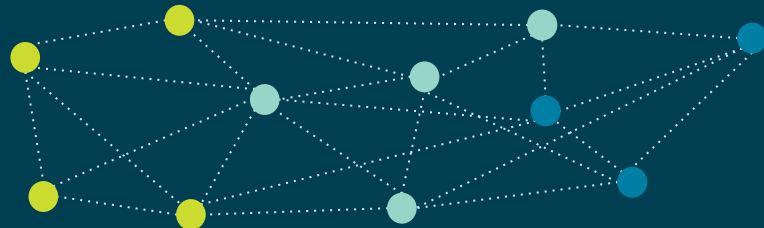
Top-Down



Bottom-Up



Enterprise
Knowledge
Graph



Retail Store Customer Order Inventory
Product

US_Customers

cid	first_name	last_name	email
100% distinct 500 records	97% distinct 500 records	99.40% distinct 500 records	100% distinct 500 records
1	Gabriello	Stocky	gstocky0@umich.edu
2	Merna	Fruchter	mfruchter1@g.co
3	Nelly	Bloys	nbloys2@nps.gov
4	Orren	Adne	oadne3@goo.gl

Products

id	product	msrp	dept
8.79% distinct 80 records		99.83% distinct 580 records	4.66% distinct 579 records
	Ring Light	568.66	8
	Wireless Charger	275.17	8
	Notebook	492.14	23
	Snowglobe	589.83	21



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Ways to Start

- Create the data model by hand (**Top-Down**)
- Use the data (CSVs or SQL DBs) to inform the data model (**Bottom-Up**)
- Load the data model from another Designer project
- Import the data model from an existing Stardog database
- Load the data model from an Ontology editor (e.g. Protégé)
- Load an open source ontology



Naming conventions

Pick a convention for naming things - any convention - and stick with it.

Entity and Relationship Naming Conventions:

- **Entity Names:**
 - a. Use **singular form** for class names (e.g., *Customer*, *Product*, *Purchase*), as it represents a single instance of the entity.
- **Relationship Names:**
 - a. Use **camelCase**, with the verb in the **present tense**, or preposition followed by the destination class (e.g., *buysProduct* for the relationship between *Customer* and *Product*).
- **Attribute Names:**
 - a. Avoid verbs, and describe them in a **concise and descriptive** manner (e.g., *Age*, *Price*, *Name*).

*Avoid unnecessary prefixes or abbreviations to keep names intuitive.

Data Modeling Terminology



STARDOG

**Stardog is an Enterprise Knowledge Graph
based on RDF.**



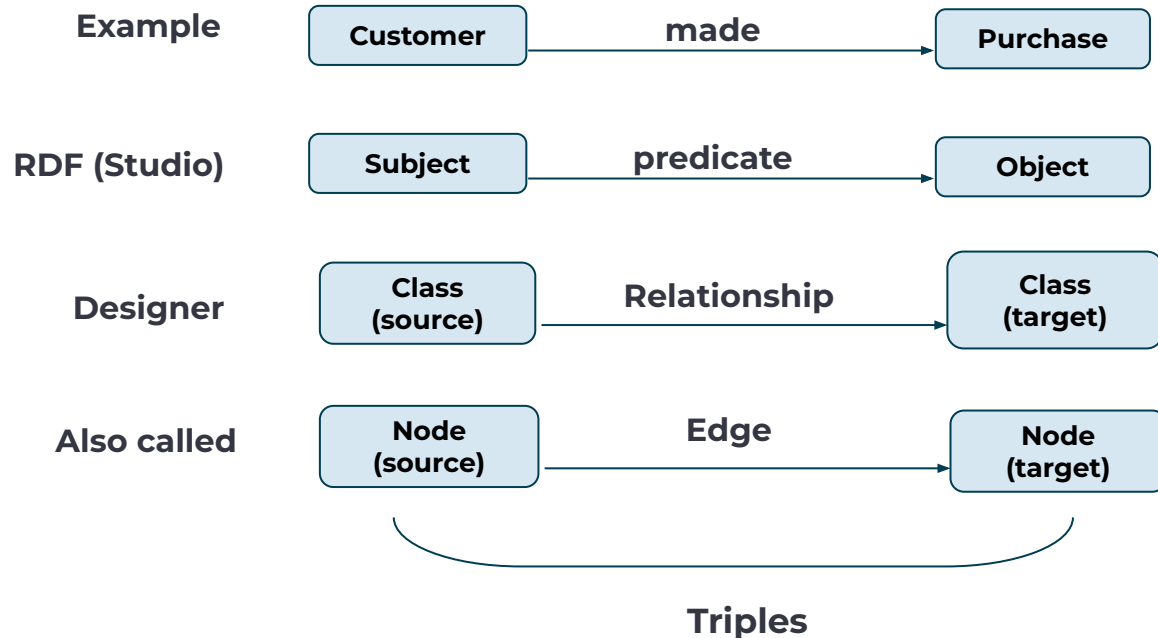
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RDF Triple & Graph

An **RDF triple** is made up of a subject, predicate and object.

An **RDF graph** is a set of triples.

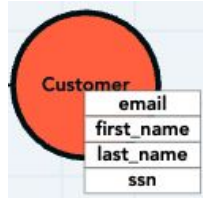
Note: In Designer, some of the naming conventions differ from the traditional RDF terms.



Modeling Terminology



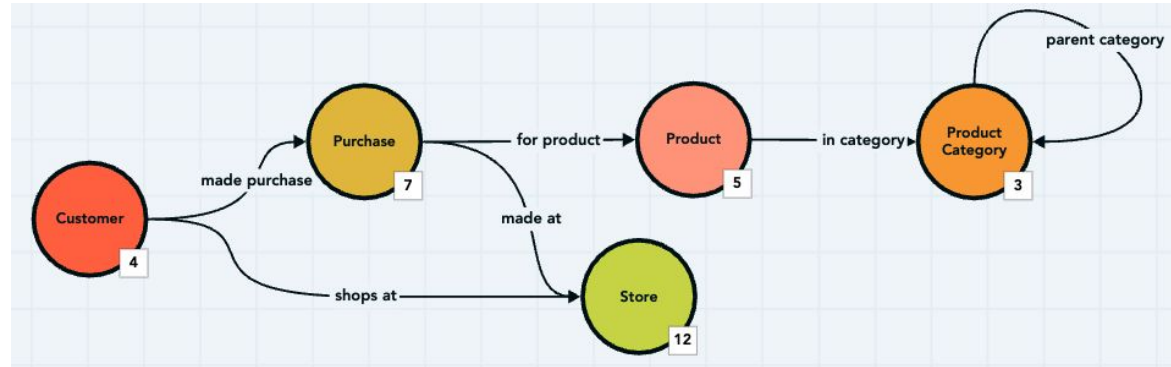
Class is a type of thing. A class can be made up of a set of individuals which are called instances.



Class with 3 attributes. **Attributes** describe properties of class instances.

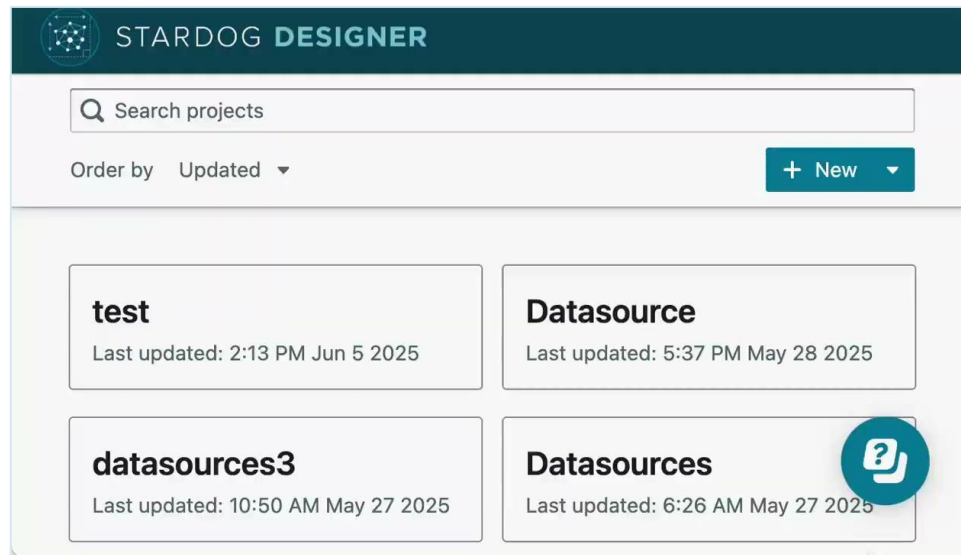


Relationships describe an edge between 2 objects (classes or individuals).



Saving Projects

- Projects can now be promoted by users, saving them to cloud storage or, if on-premise, to the account.
- Each project is specific to its account and endpoint.
- For local storage, projects will be saved within the user's browser.
- To share a project, export it as a .zip file or commit it to a source control system.



Designer Canvas Legend



Resource
(file or table)



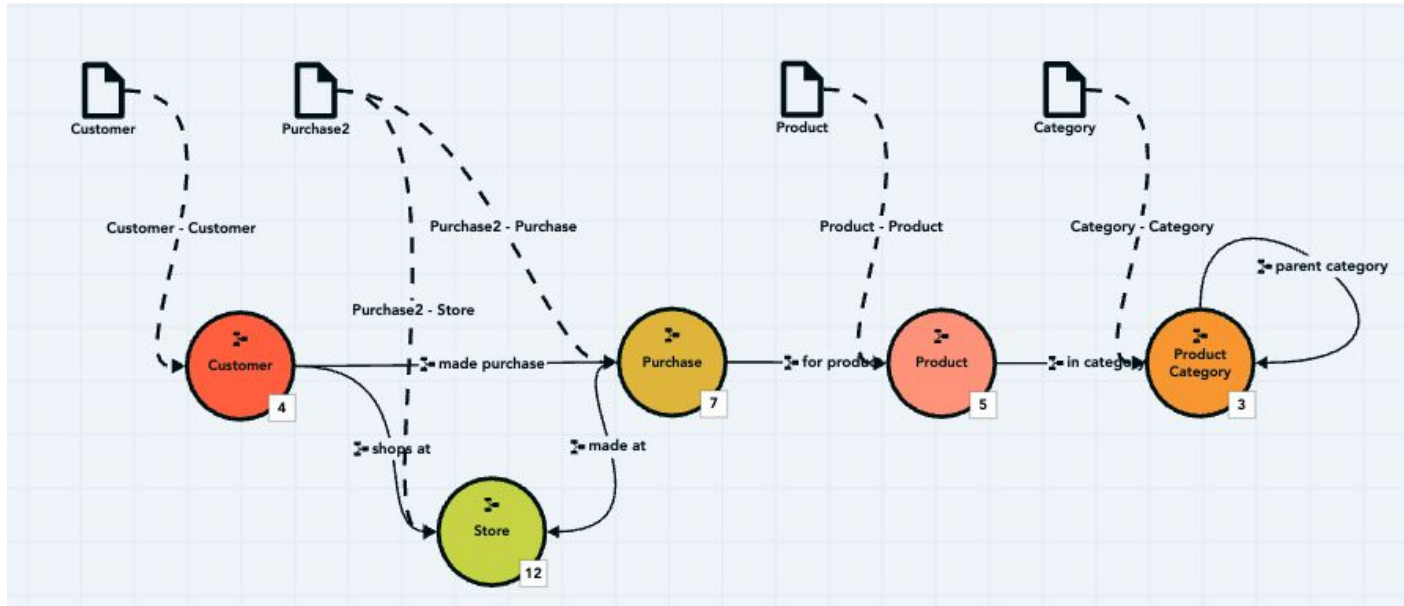
Mapping
(dotted line)



Mapped Class



Mapped Relationship



Stardog Designer Model User Interface

The image shows the Stardog Designer Model User Interface with several callouts explaining its features:

- Designer Training**: The main title of the application.
- Search**: A search bar in the top left.
- Questions 0**: A section for questions, currently showing 0.
- Resources 0**: A section for resources, currently showing 0.
- Specify data to be mapped**: A callout pointing to the Resources section.
- Data Model**: A tab in the top navigation bar.
- Mapping**: A tab in the top navigation bar.
- Toggle between Model overview and Mapping pages**: A callout pointing to the tabs.
- Model: Designer Training**: The current model being edited.
- Import data models from Stardog endpoints**: A callout pointing to the top left of the main workspace.
- Redraw layout**: A callout pointing to the top center of the main workspace.
- Create a class**: A callout pointing to the top center of the main workspace.
- Delete concept**: A callout pointing to the top center of the main workspace.
- Create a relationship**: A callout pointing to the top center of the main workspace.
- Manage rules**: A callout pointing to the top right of the main workspace.
- Flip between Tabular / Graph view**: A callout pointing to the top right of the main workspace.
- Publish project to a Stardog endpoint**: A callout pointing to the top right of the main workspace.
- Open Voicebox**: A callout pointing to the bottom left of the main workspace.

Tabular View of Data Model

The tabular view shows Classes, Relationships and Attributes defined in your Designer project.



Classes Relationships Attributes		
Columns Q Search		
Class Name	Attributes	KG Training...
Category	3	●
Customer	4	●
Product	5	●
Purchases	7	●

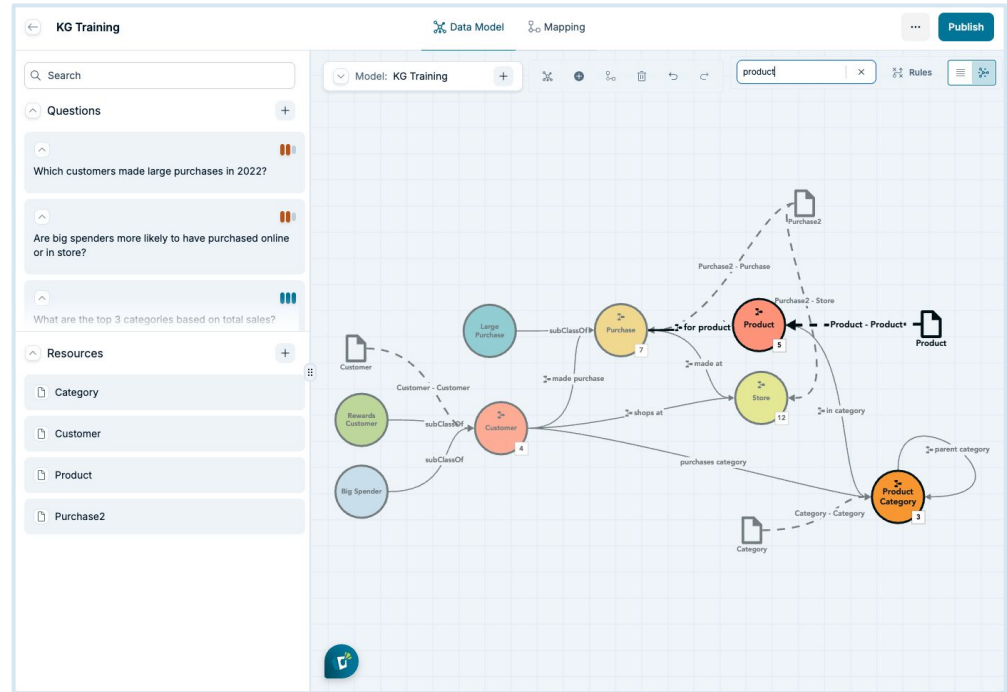
Classes Relationships Attributes			
Columns Q Search			
Relationship Name	Sources	Targets	KG Training...
for product	Purchases	Product	●
in category	Product	Category	●
made purchase	Customer	Purchases	●
parent category	Category	Category	●

Classes Relationships Attributes			
Columns Q Search			
Attribute Name	Data Types	Associated Clas...	KG Training...
card	string	Purchases	●
cid	integer	Purchases	●
date	date	Purchases	●
dept	integer	Product	●
dept_name	string	Category	●
desc	string	Product	●
description	string	Category	●
email	string	Customer	●
first_name	string	Customer	●
id	integer		●
last_name	string	Customer	●
msrp	float	Product	●
parent	string	Category	●
pid	integer	Purchases	●
price	float	Purchases	●
product	string	Product	●
quantity	integer	Purchases	●
reward_account	string	Purchases	●
ssn	string	Customer	●



Search Across Canvas

The tabular view shows Classes, Relationships and Attributes defined in your Designer project.



Edit Classes

- Edit label and description
- Customize IRI
- Manage parent and child classes
- Add relationships
- Manage attributes
- Set Geospatial visualization parameters
- Delete class



The screenshot shows a 'Class Detail' form with a close button (X) in the top right corner. Below the title is a horizontal tab bar with five tabs: 'Basic' (selected), 'Hierarchy', 'Relationships', 'Attributes', and 'Geospatial'. The 'Basic' tab contains two input fields: 'Class Name' with the value 'Purchase' and a yellow dot on the right, and 'Description' with an empty text area. At the bottom of the form is a 'Delete' button.

Edit Relationships

- Edit label and description
- Customize IRI
- Manage source and target classes
- Delete relationship

Relationship Detail

Relationship Name

made purchase

Description

Source

Customer

x

Target

Purchase

x

Source	Target
Customer	Purchase

Delete

Edit Attributes

- Edit label and description
- Customize IRI
- Edit Data Type
- Manage class associations
- Manage sensitive property group member
- Delete attribute

The screenshot shows a web interface for editing an attribute. The form is titled 'Attribute Detail' and includes the following sections:

- Attribute Name:** A text input field containing the value 'email'.
- Description:** A large text area for providing a description.
- Data Type:** A dropdown menu currently set to 'string'.
- Associations:** A dropdown menu with 'Customer' selected and a delete icon (x) next to it.
- Sensitive Property Groups:** A section with a '+ Create' button and a search filter input labeled 'Filter'. Below this is a table with two columns: 'Name' and 'Other attributes'.

A 'Delete' button is located at the bottom left of the form.

The Store Use Case

Purchase

id	pid	date	time	quantity	price	cid	card	reward_acc	Store_Type	StoreID	Store_URL	Store_Address	Store_City	Store_Zip	Store_State
bad78090-e8	176	8/8/19	7:45 AM	11	9722.24	206	fcc08543-ef9e-42a9-b6ac		Online	1	https://store.example.com				
9ebf4c10-95	220	4/19/19	12:35 PM	19	11941.69	18	44f8a099-74c3f53240-13		Online	1	https://store.example.com				
20578c3c-1b	237	7/2/21	4:48 AM	15	11397.45	257	ea40b1a1-7437483d4d-57		Online	1	https://store.example.com				
f69b0158-58	522	8/16/12	10:56 PM	22	16439.94	342	ca34a80a-e360555aef-3a		Online	1	https://store.example.com				
93b06e51-bc	183	5/20/15	5:55 PM	25	18927.75	278	762a0179-9e81eb3dd1-55		Physical Store	7		6145 Main St	San Jose	95101	CA
beed44b7-fd	160	5/15/15	6:38 PM	8	2177.84	220	5878519f-f679-47fb-9702-		Physical Store	2		1242 Main St	New York	10001	NY

Store

US_Customers

cid	first_name	last_name	email
100% distinct 500 records	97% distinct 500 records	99.40% distinct 500 records	100% distinct 500 records
1	Gabriello	Stocky	gstocky0@umich.edu
2	Merna	Fruchter	mfruchter1@g.co
3	Nelly	Bloys	nbloys2@nps.gov
4	Orren	Adne	oadne3@goo.gl

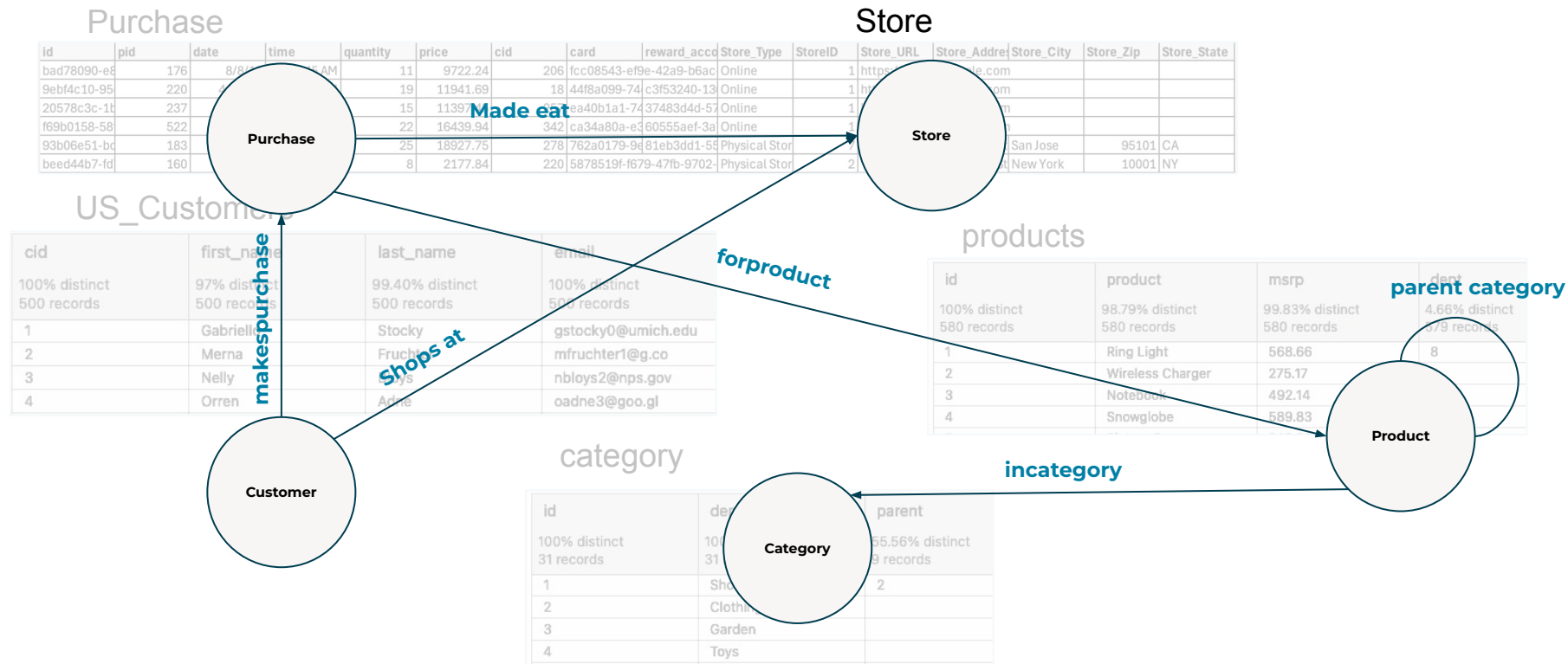
products

id	product	msrp	dept
100% distinct 580 records	98.79% distinct 580 records	99.83% distinct 580 records	4.66% distinct 579 records
1	Ring Light	568.66	8
2	Wireless Charger	275.17	8
3	Notebook	492.14	23
4	Snowglobe	589.83	21

category

id	dept_name	parent
100% distinct 31 records	100% distinct 31 records	55.56% distinct 9 records
1	Shoes	2
2	Clothing	
3	Garden	
4	Toys	

The Store Use Case



Exercise



- Open the Exercise Guide
- Complete Exercises 1, 2, & 3

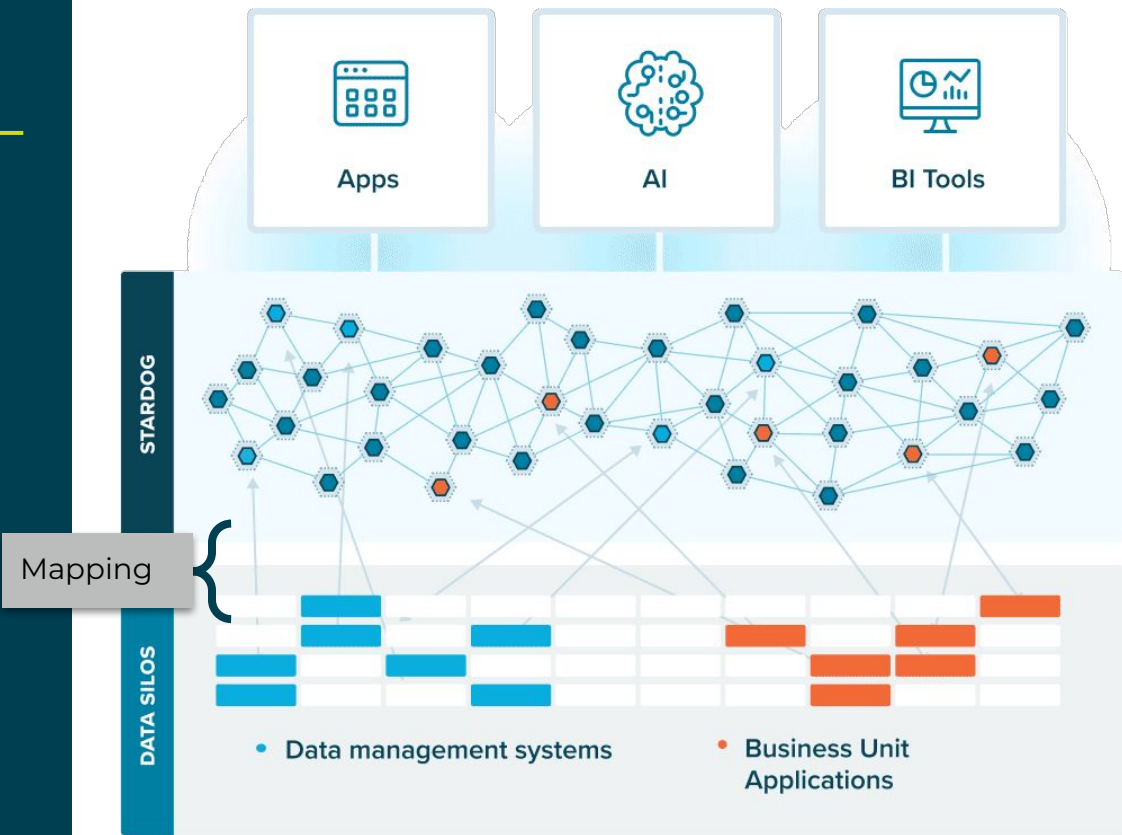
Mapping



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Where Mapping sits

- Mapping happens at the junction between the 2 worlds of RDB and Graph
- In your data you have
 - Files and Tables
 - Column
 - JoinsThat will become or be “mapped to”
 - Classes
 - Attributes
 - RelationshipsIn the graph world

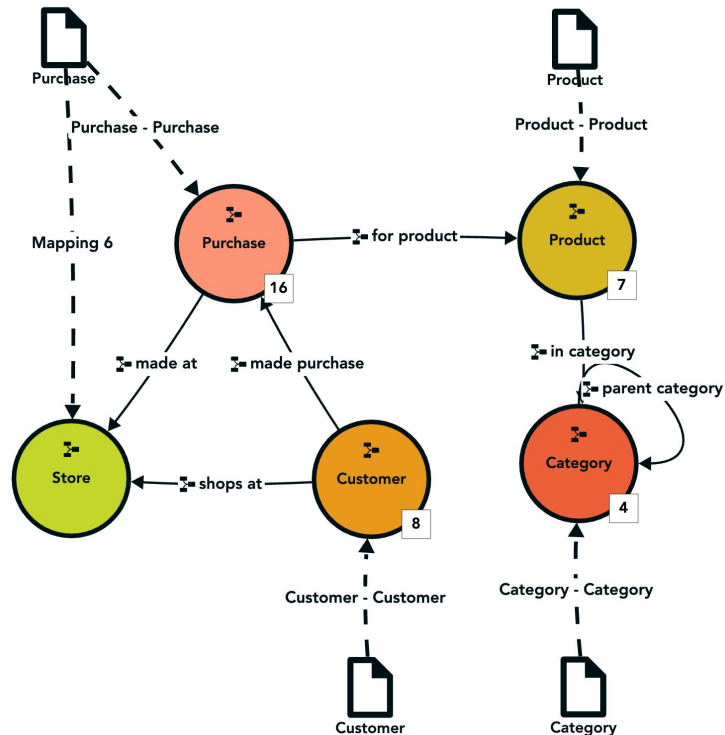


Mapping Project Resources

Designer facilitates mapping new data sources to Stardog.

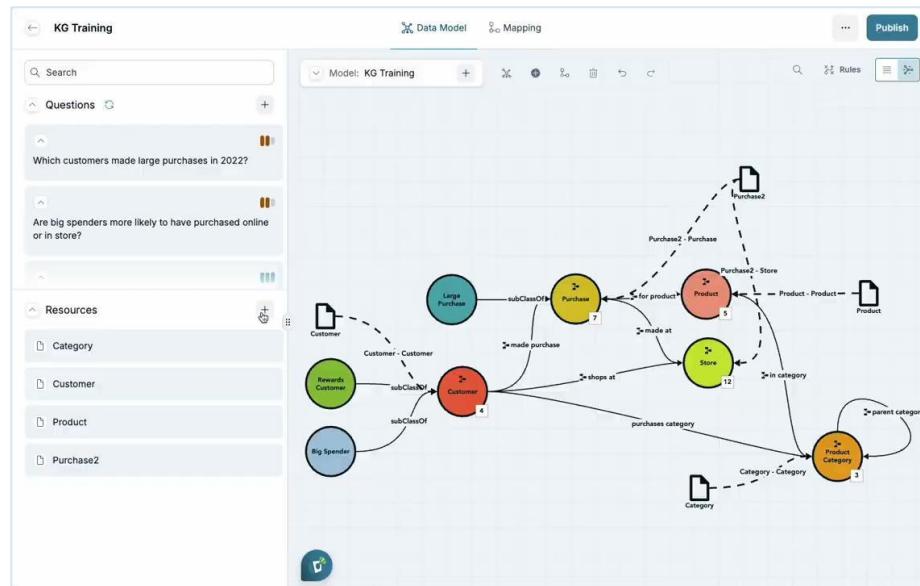
Supported data sources include:

- CSVs
- JDBC Data Connections



CSVs

- Rename resource and column headers
- See summary statistics and data preview
- Customize separator, quote, and escape characters
- Upload multiple CSV resources at the same time



Edit CSV Resources

- Edit resource name
- Edit resource property
- Mappings are preserved when possible
- Replace CSV with a new file

File Upload Preview

Source

products.csv

Replace

Resource name

products

Resource Properties

Separator character

,

Quote character

"

Escape character

"

Has header

Skip empty values

Preview 25 samples

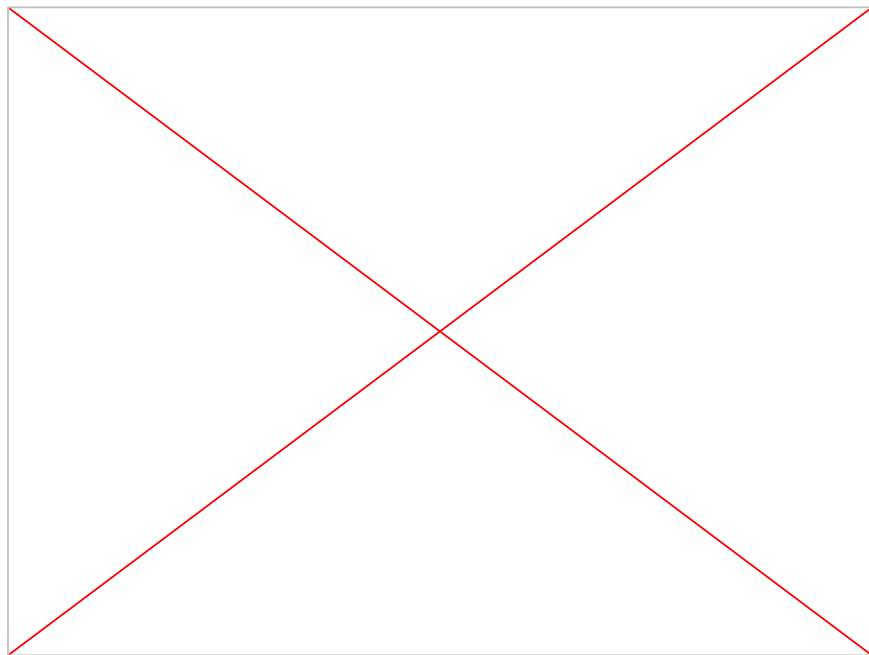
productID	productName	quantityPerUnit	unitPrice	discontinued	categoryID
100% distinct 77 records	100% distinct 77 records	90.91% distinct 77 records	80.52% distinct 77 records	2.60% distinct 77 records	10.39% distinct 77 records
1	Chai	10 boxes x 20 bags	18	0	1
2	Chang	24 - 12 oz bottles	19	0	1
3	Aniseed Syrup	12 - 550 ml bottles	10	0	2
4	Chef Anton's Cajun S...	48 - 6 oz jars	22	0	2
5	Chef Anton's Gumbo...	36 boxes	21.35	1	2
6	Grandma's Boysenbe...	12 - 8 oz jars	25	0	2
7	Uncle Bob's Organic ...	12 - 1 lb pkgs.	30	0	7
8	Northwoods Cranber...	12 - 12 oz jars	40	0	2
9	Mishi Kobe Niku	18 - 500 g pkgs.	97	1	6
10	Ikura	12 - 200 ml jars	31	0	8
11	Queso Cabrales	1 kg pkg.	21	0	4
12	Queso Manchego La ...	10 - 500 g pkgs.	38	0	4
13	Konbu	2 kg box	6	0	8
14	Tofu	40 - 100 g pkgs.	23.25	0	7
15	Genen Shouyu	24 - 250 ml bottles	15.5	0	2
16	Pavlova	32 - 500 g boxes	17.45	0	3

Cancel

Save

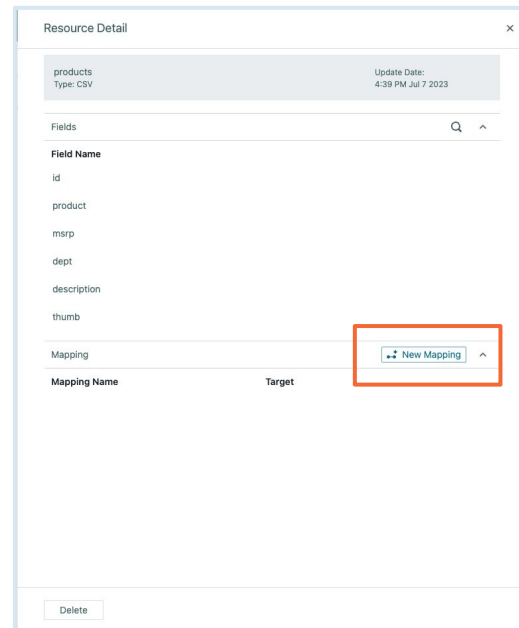
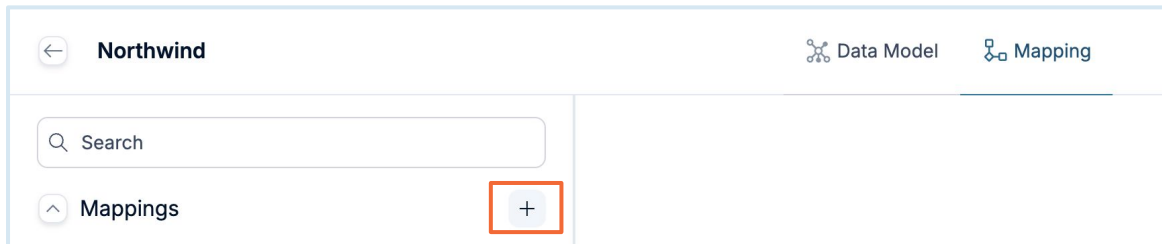
Virtual Graphs via JDBC Data Connections

- Select database tables or select data with a custom query
- Rename resource and column headers for VGs and CSV
- Includes Data preview



Add mapping

- **New Mapping** button on resource details
- **New Mapping** button on Mapping view



Stardog Designer Mapping User Interface

The screenshot displays the Stardog Designer Mapping User Interface. On the left is a navigation tree showing project resources, classes, and mappings. The main area is divided into two tabs: 'Data Model' and 'Mapping'. The 'Mapping' tab is active, showing a mapping configuration for 'Category - C...' with a 'Product Cate...' category. The mapping is configured with a 'Primary Identifier' of 'id' and a 'Label' of 'dept_name'. The 'Relationships' section shows three relationships: 'in category', 'parent category', and 'purchases category'. The 'Attributes' section shows three attributes: 'dept_name', 'description', and 'parent'. A 'Publish' button is visible in the top right corner.

Create new mappings

Toggle between Model overview and Mapping pages

Navigation tree of project resource, class, and mappings

Mapping Configuration:

Category	Type
Category - C...	CSV

Primary Identifier

Identifier
id

Label

Label
dept_name

Relationships

Relationship	Source Class	Target Class	Target Identifier
in category			
parent category	Product Category	parent	
purchases category			

Attributes

Attribute	Type	Value
dept_name	string	dept_name
description	string	description
parent	string	parent

Exercise



- Complete Exercise 4 - Upload Your Data
 - Upload and map data
- Complete Exercise 5 - Edit Classes

Relationships



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The Store Use Case

Purchase

id	pid	te	time	quantity	price	cid	card	reward_account	Store_Type	Store_URL	Store_Address	Store_City	Store_Zip	Store_State
bad78090-e88e-	176	8/8/19	7:45 AM	11	9722.24	206	fcc08543-ef9e-42a9-b6ac-5b913df		Online	https://store0.example.com				
9ebf4c10-95de-4	220	4/19/19	12:35 PM	19	11941.69	18	44f8a099-74c3f53240-136c-4e0b-		Online	https://store1.example.com				
20578c3c-1b31-	237	7/2/21	4:48 AM	15	11397.45	257	ea40b1a1-7437483d4d-5758-468b-		Online	https://store2.example.com				
f69b0158-58f4-4	522	8/16/12	10:56 PM	22	16439.54	342	ca34a80a-e360555aef-3a70-4216-		Online	https://store3.example.com				
93b06e51-bc44-	183	5/20/15	5:55 PM	25	18927.75	278	762a0179-9d81eb3dd1-5583-4785-		Physical Store		6145 Main St	San Jose	95101	CA

US_Customers

cid	first_name	last_name	email
100% distinct 500 records	97% distinct 500 records	99.40% distinct 500 records	100% distinct 500 records
1	Gabriello	Stocky	gstocky0@umich.edu
2	Merna	Fruchter	mfruchter1@g.co
3	Nelly	Bloys	nbloys2@nps.gov
4	Orren	Adne	oadne3@goo.gl

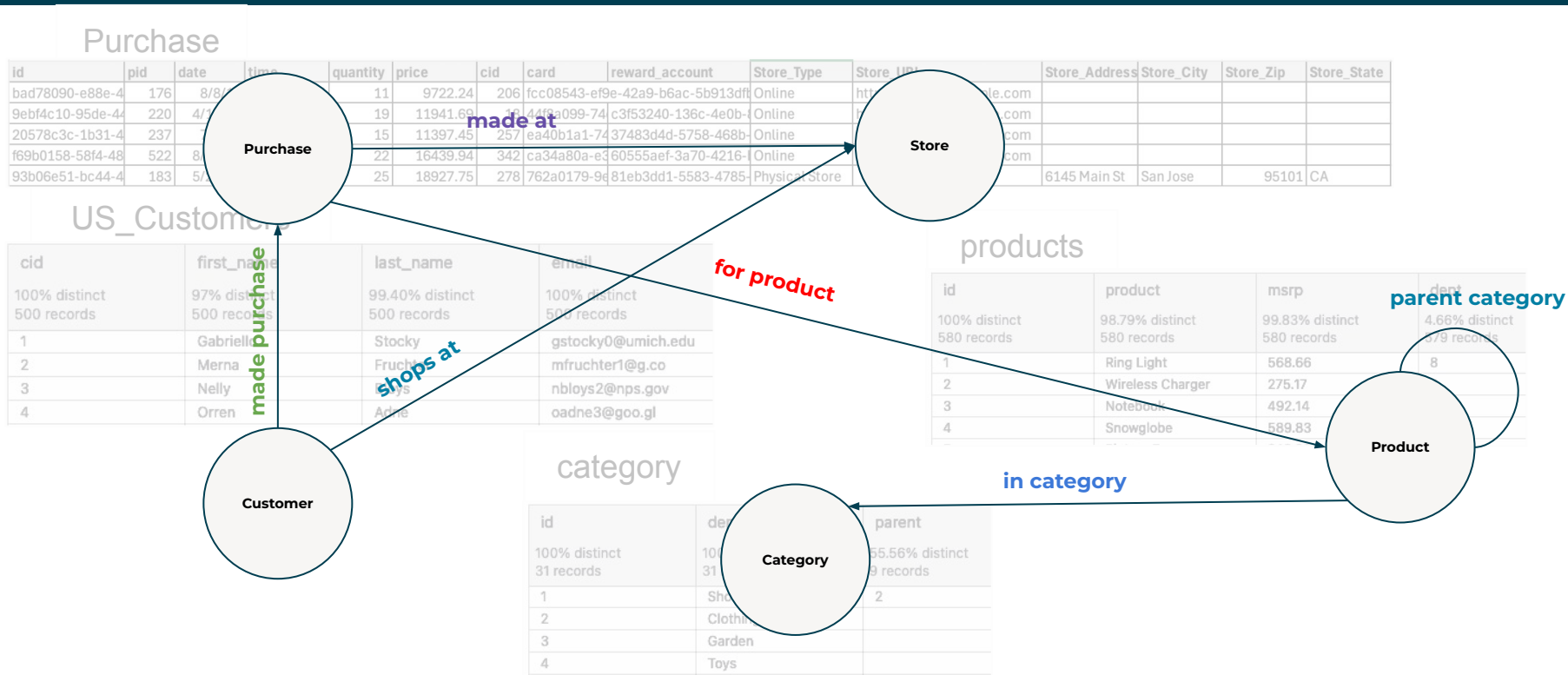
products

id	product	msrp	dept
100% distinct 580 records	98.79% distinct 580 records	99.83% distinct 580 records	4.66% distinct 579 records
1	Ring Light	568.66	8
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category

id	dept_name	parent
100% distinct 31 records	100% distinct 31 records	55.56% distinct 9 records
1	Shoes	2
2	Clothing	
3	Garden	
4	Toys	

The Store Use Case



Map the Relationship

Map project resources to relationships.

- Mapping Toggle
- Fill in the target class & target identifier under relationships
- Version 11.1+: map to a static string

The screenshot displays the 'KG Training' mapping interface. On the left, a tree view shows the project structure: Mappings, Customer (Customer - Customer), Product (Product - Product), and Purchase2 (Purchase, Purchase2 - Purchase, Store, Purchase2 - Store). The 'Purchase2 - Purchase' relationship is selected. The main panel shows the mapping configuration for 'Purchase2' (Type: CSV). It includes fields for Primary Identifier (set to 'id'), Label (set to 'id'), and Relationships. The 'made at' relationship is configured with Target Class 'Store' and Target Identifier '1'. The 'made purchase' relationship is configured with Source Class 'Customer' and Source Identifier 'cid'. The bottom section shows attributes: 'card' (string), 'cid' (integer), and 'date' (date).

Relationship	Target Class	Target Identifier	Preview
for product	Product	pid	176
made at	Store	1	urn:KG_Training:Store: 1
made purchase	Customer	cid	206

Attribute	Type	Value
card	string	card
cid	integer	cid
date	date	date

Exercise

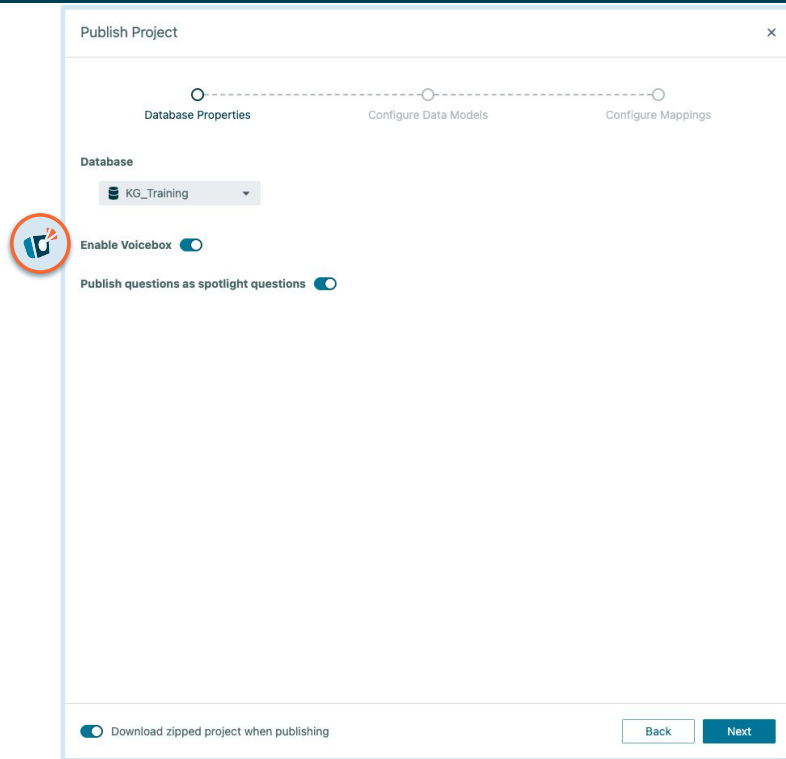


- Complete Exercise 6 - Create and Map Relationships
- Review

Publish the Model

Database

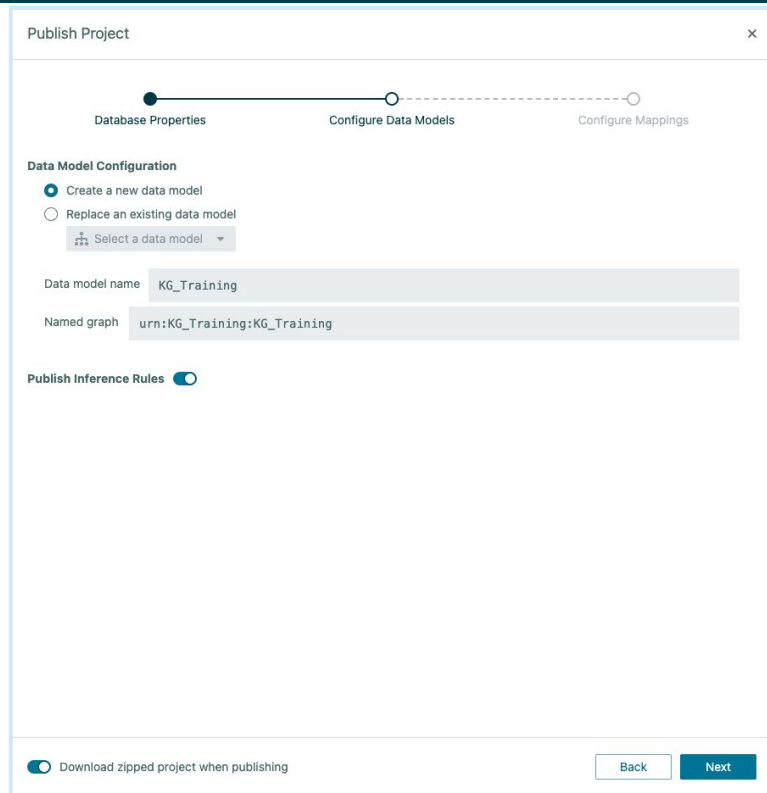
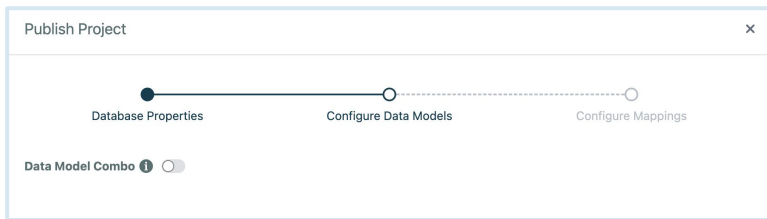
- Select or create a database
- Download project files
 - Same files as the export option



The screenshot shows a 'Publish Project' dialog box with a progress bar at the top containing three steps: 'Database Properties' (active), 'Configure Data Models', and 'Configure Mappings'. Below the progress bar, the 'Database' section shows a dropdown menu with 'KG_Training' selected. To the left of this section is a red circle containing the Microsoft logo icon. Below the database dropdown are two toggle switches: 'Enable Voicebox' (checked) and 'Publish questions as spotlight questions' (checked). At the bottom of the dialog, there is a checkbox labeled 'Download zipped project when publishing' (checked) and two buttons: 'Back' and 'Next'.

Configure Data Models

- Create or Replace the data model
- Edit data model name
- Edit Named Graph
- (Optional) Publish Data Model Combos



Configure Mappings

- Choose how to publish resources
 - Create new
 - Replace
 - Append
- Customize adv resource details
- Publish

The screenshot shows the 'Publish Project' dialog box with the 'Configure Mappings' step selected in the progress bar. The dialog contains a table with columns: Publish, Resource, Mappings, and Configuration. There are four rows of resources: Customer, Product, Category, and Purchase2. Each row has a 'Publish' checkbox (all checked), a 'Resource' name, a 'Mappings' name, and a 'Configuration' section with radio buttons for 'Create new' (selected), 'Replace', and 'Append'. Below the table is a 'Named graph' field with the value 'urn:KG_Training:Customer'. At the bottom, there is a checkbox for 'Download zipped project when publishing' (unchecked) and 'Back' and 'Publish' buttons.

Publish	Resource	Mappings	Configuration
☒	Customer	Customer - Customer	☒ Create new ☐ Replace ☐ Append
☒	Product	Product - Product	☒ Create new ☐ Replace ☐ Append
☒	Category	Category - Category	☒ Create new ☐ Replace ☐ Append
☒	Purchase2	Purchase2 - Purchase Purchase2 - Store	☒ Create new ☐ Replace ☐ Append

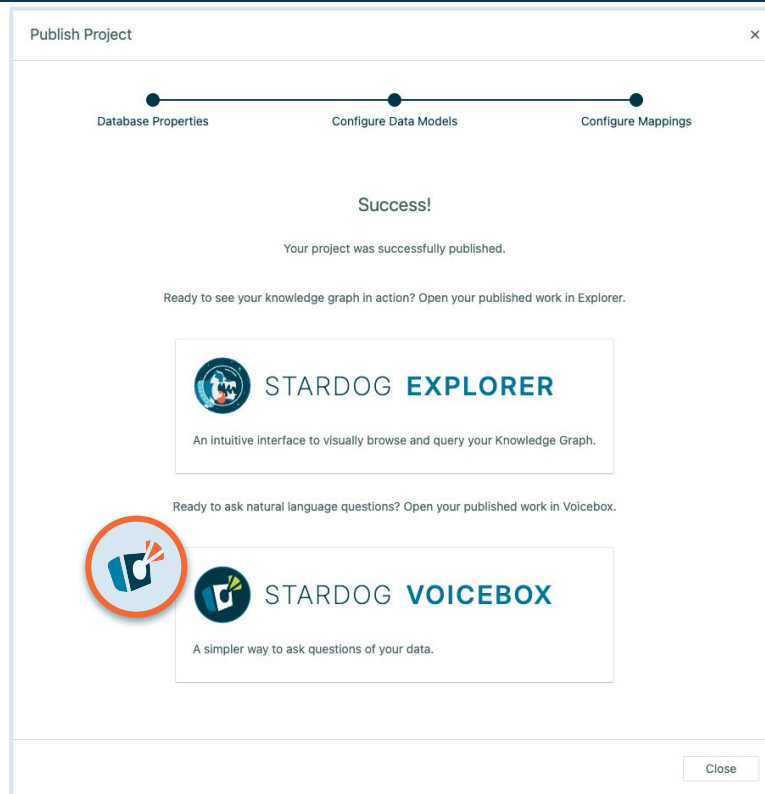
Named graph: urn:KG_Training:Customer

☐ Download zipped project when publishing

Back Publish

Success!

- Exported project files downloaded locally if option was checked
- Launch Applications
 - Stardog Explorer
 - Stardog Voicebox

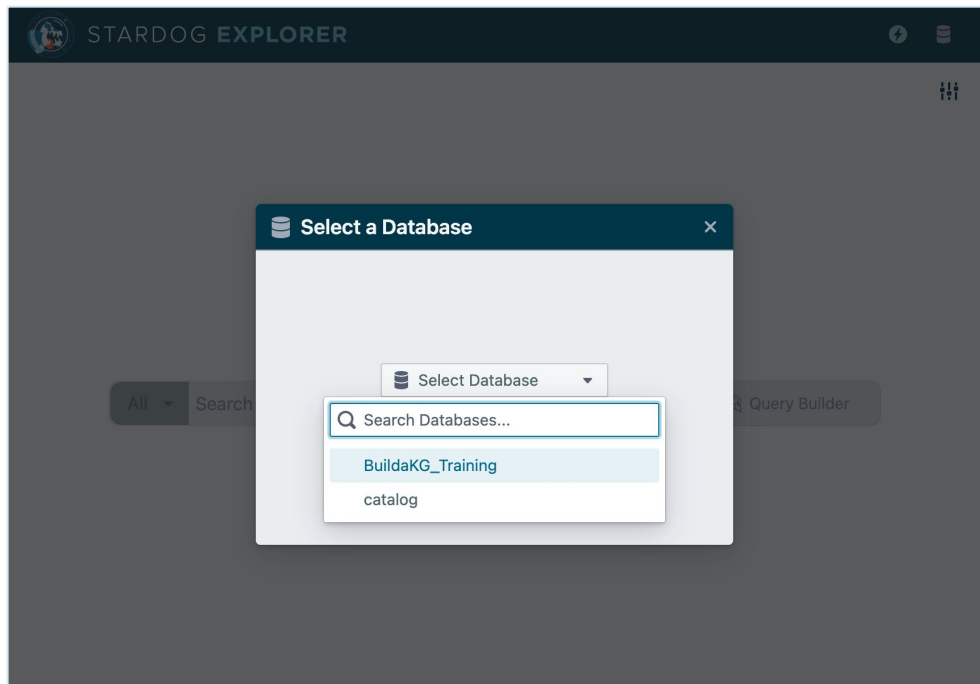


Project Export

- Data Model
 - In Turtle (.ttl) format
- Project Resource mappings
 - Mapping file in Stardog Mapping Syntax (SMS) format
 - Properties file
 - (For CSV resources) Original CSV
- Stardog Designer Project definition
 - With extension `.stardogdesigner`

Validate Model in Explorer

- Go to <https://cloud.stardog.com>
- Open Explorer
- Select database
- Select visualize

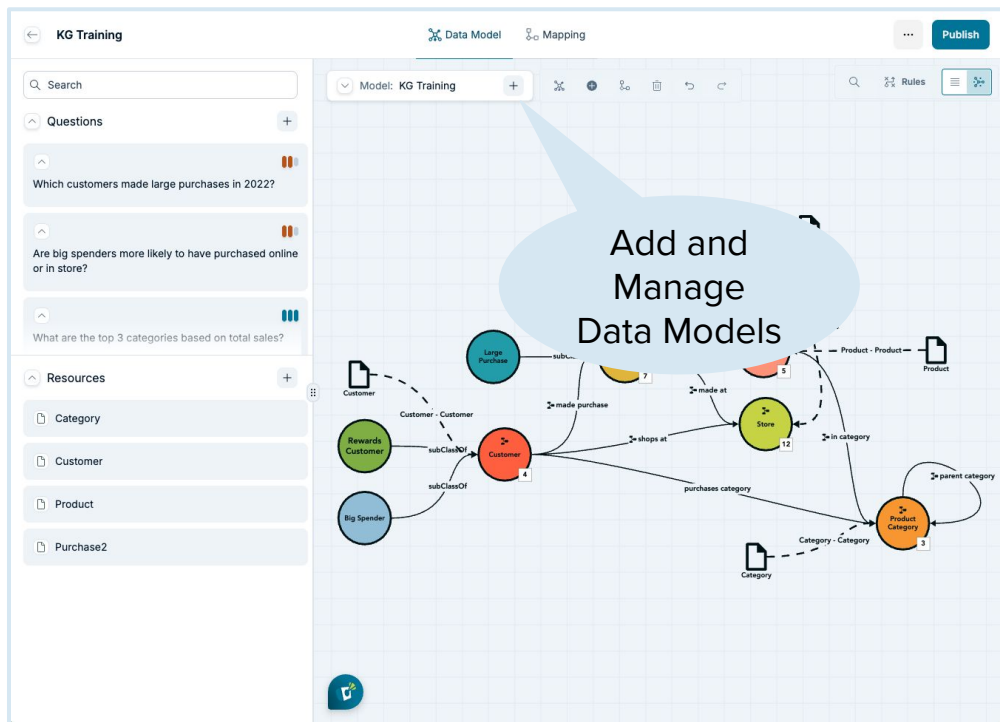


Exercise

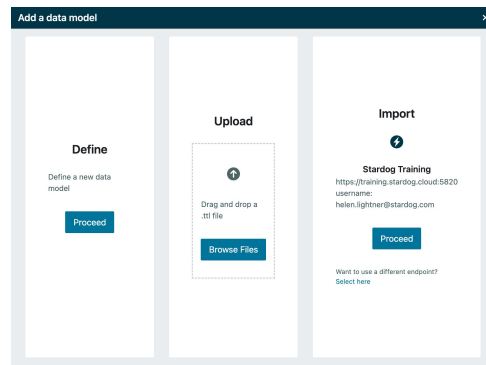


- Complete Exercise 7 - Publish your project
- Review

Managing Multiple Models in Designer



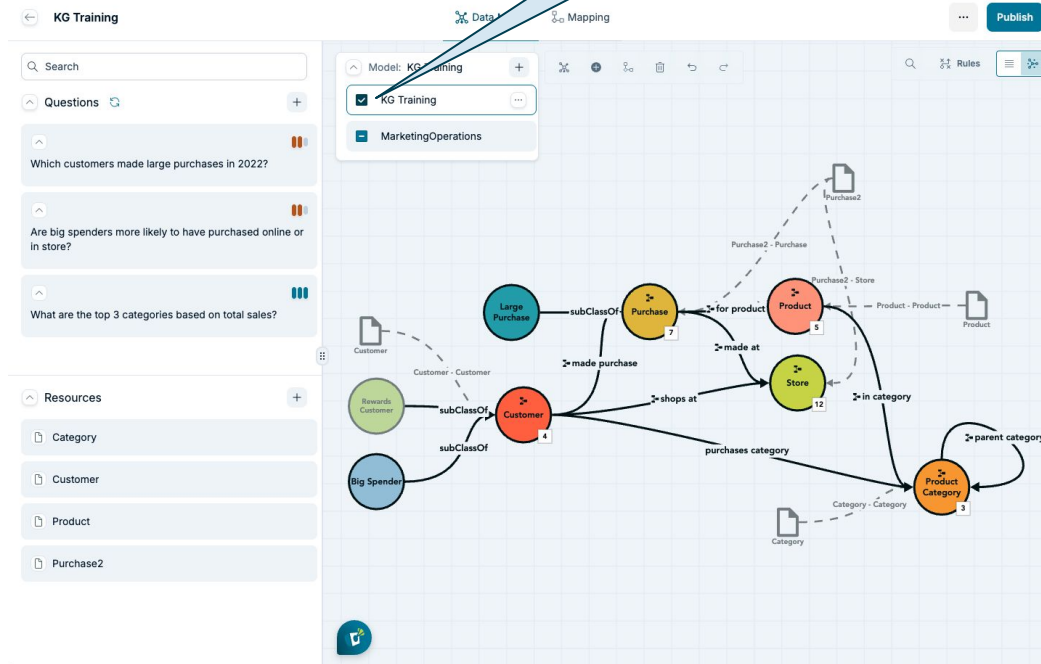
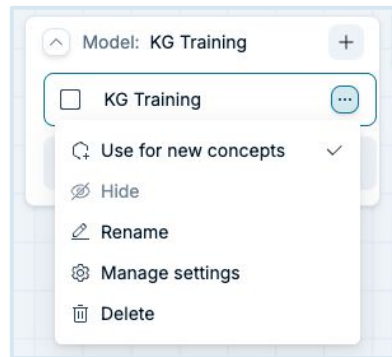
- Add a data model by building a new one or importing one from another database or a .ttl



Working with Multiple Data Models

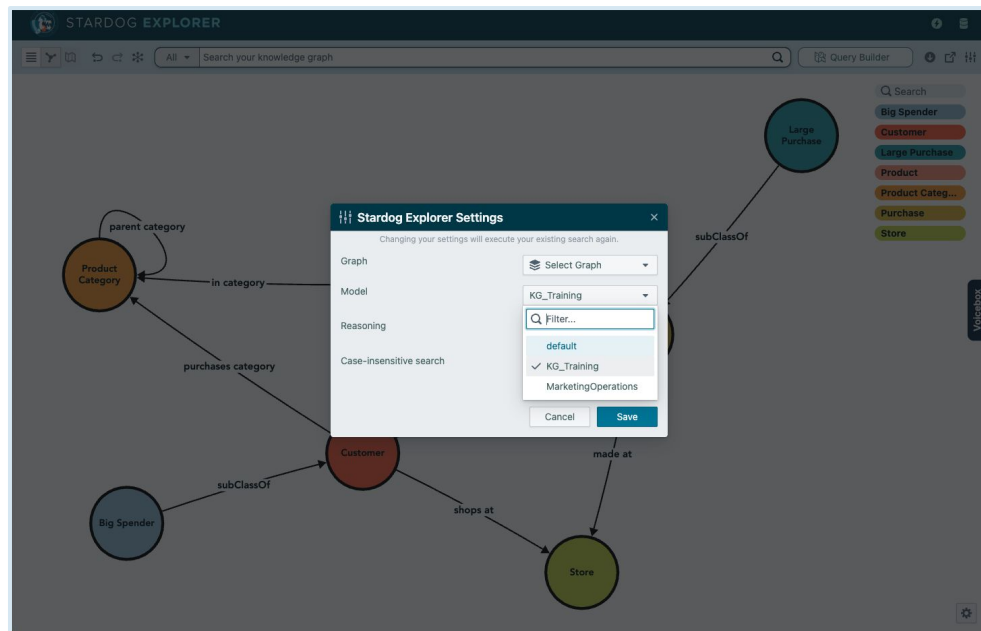
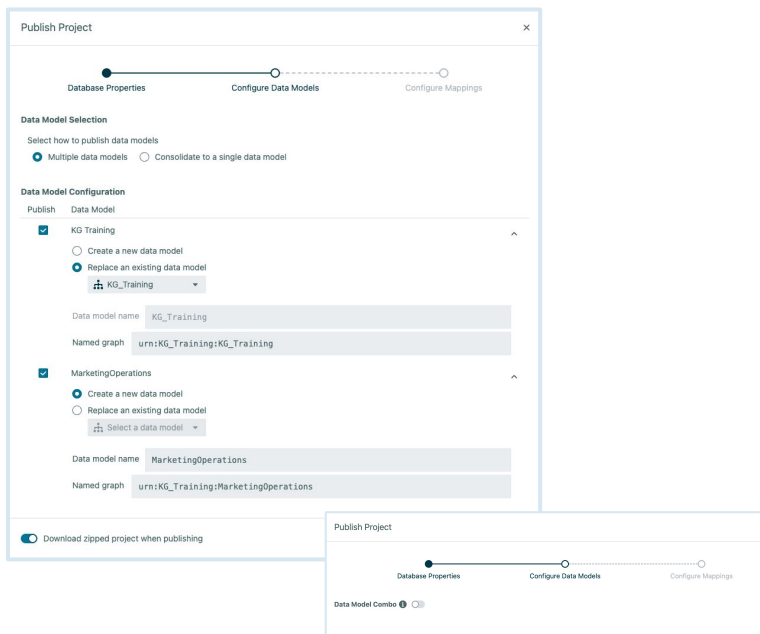
- Add multiple data models to the canvas
- ... data model options

Click the check boxes to see which concepts are a part of a specific data model



Publishing Multiple Data Models (Designer)

- Publish multiple data models
- Both models can be accessed in Settings under Model



Publishing Multiple Data Models (Designer)

- Publish multiple data models

Publish Project

Database Properties Configure Data Models Configure Mappings

Data Model Selection

Select how to publish data models

☒ Multiple data models ☐ Consolidate to a single data model

Data Model Configuration

Publish **Data Model**

☒ KG Training

☐ Create a new data model

☒ Replace an existing data model

KG_Training

Data model name: KG_Training

Named graph: urn:KG_Training:KG_Training

☒ MarketingOperations

☐ Create a new data model

☐ Replace an existing data model

Select a data model

Data model name: MarketingOperations

Named graph: urn:KG_Training:MarketingOperations

☒ Download zipped project when publishing

Back Next

- Combo data model: links 1 or more data models using a single identifier
- Create a new or replace an existing data model
- Select / Deselect data models to include

Publish Project

Database Properties Configure Data Models Configure Mappings

Data Model Combo

☒ Create a new data model

☐ Replace an existing data model

Select a data model

Data model name: KG_Training_data_model_combo

Included data model named graphs

Model	Data Model(s)
☒ Data model named graph	
☒ urn:KG_Training:KG_Training	default, KG_Training
☒ urn:KG_Training:MarketingOperations	MarketingOperations

- New combo model can be accessed in the Explorer Settings or can be added to Designer as a new data model

Stardog Explorer Settings

Changing your settings will execute your existing search again.

Graph: Select Graph

Model: KG_Training_data...

Reasoning: Filter...

Case-insensitive search: ☐

default

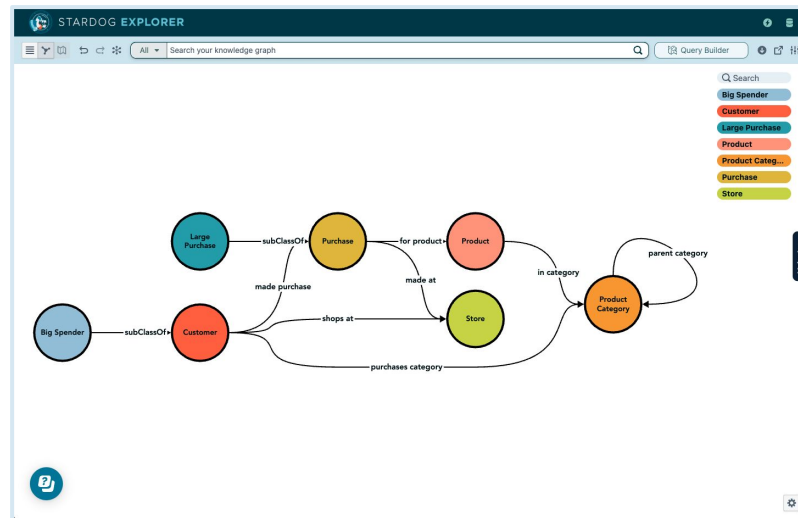
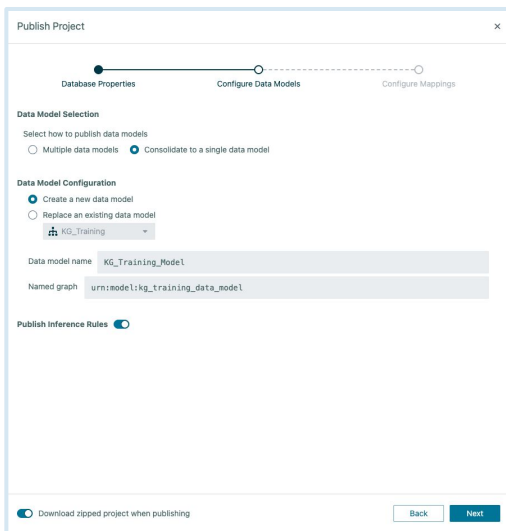
KG_Training

✓ KG_Training_data_m...

Marketin KG_Training_data_model_combo

Publishing Multiple Data Models (Designer)

- Consolidate to one data model
 - Combines selected data models into one new model
 - A new model is created or an existing one is replaced
- Access new model in Explorer Settings



Tabular View

- View and search for Classes, Relationships and Attributes in each in each data model

The screenshot shows the 'KG Training' interface in 'Tabular View'. The left sidebar contains a 'Questions' section with three queries and a 'Resources' section with four items. The main area displays a table with columns for Class Name, Attributes, and two data models: 'KG Training' and 'Marketing...'. The table lists various classes and their associated attributes and values in the two models.

Class Name	Attributes	KG Training	Marketing...
Big Spender	4	●	○
Customer	4	●	●
Large Purchase	7	●	○
Product	5	●	○
Product Category	3	●	○
Purchase	7	●	○
Rewards Customer	4	○	●
Store	12	●	○

Inference Rules

What is the power of inferencing?

Differentiates Graph from Knowledge Graph

- Make the implicit explicit
- Enables a Semantic Layer
- Formally define your business logic, **no-code**
 - Use your model to infer new edges and classes
 - Rules can be saved to one or more data models
- Inferences are computed at **query time**
 - Enables Data Fabric with Virtualization

Inference Rules

Identify statements that will be answered by applying business logic:

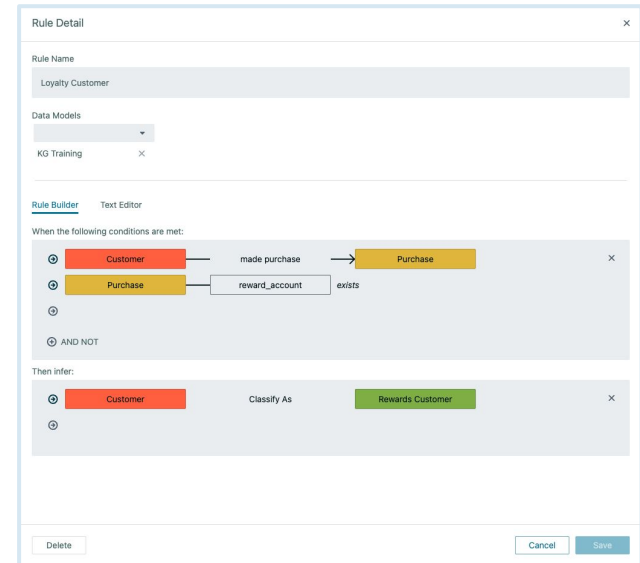
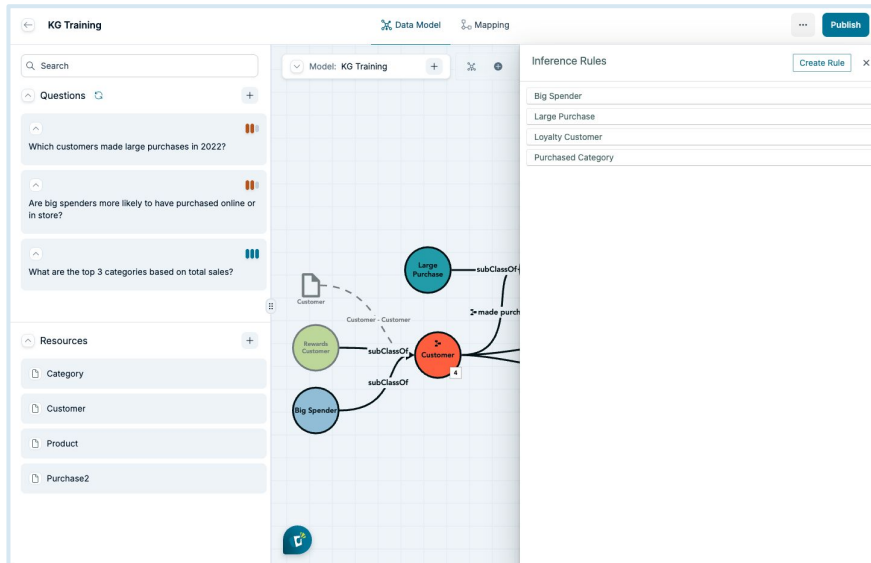
- In store shoppers are **Customers** who have an in store purchase

Rule 1: A **Customer** that makesPurchase Purchase
Purchase at a **physical store** can be classified as a Customer **In Store
Shopper**



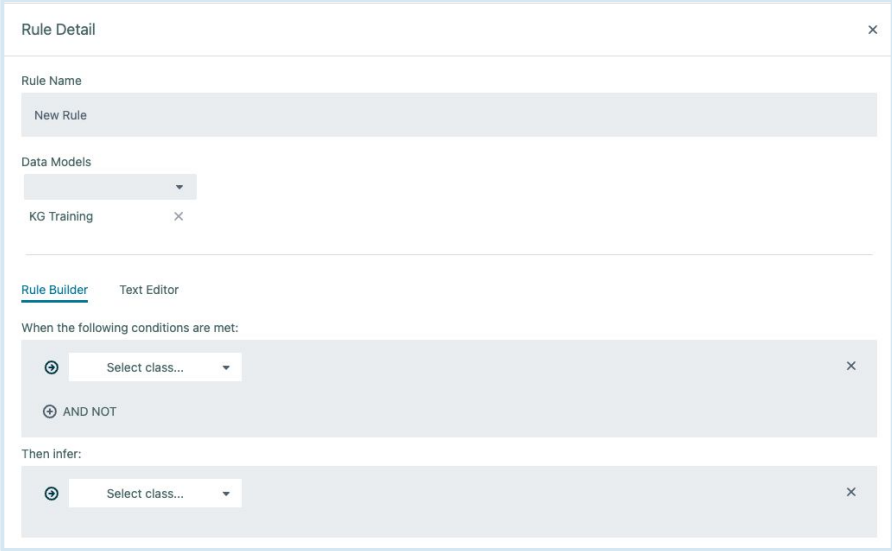
Building Rules

- View existing rules
- Build rules using a Rule Builder or a Text Editor
- Save a rule to one or more data models



Building Rules - Rule Builder

- **IF** clause specifies the conditions that are to be met. Follows the same structure as the Query Builder
- **THEN** clauses allow you
 - to classify instances as members of a class
 - to assert relationships between instances in classes
 - to assert attributes in instances of a class



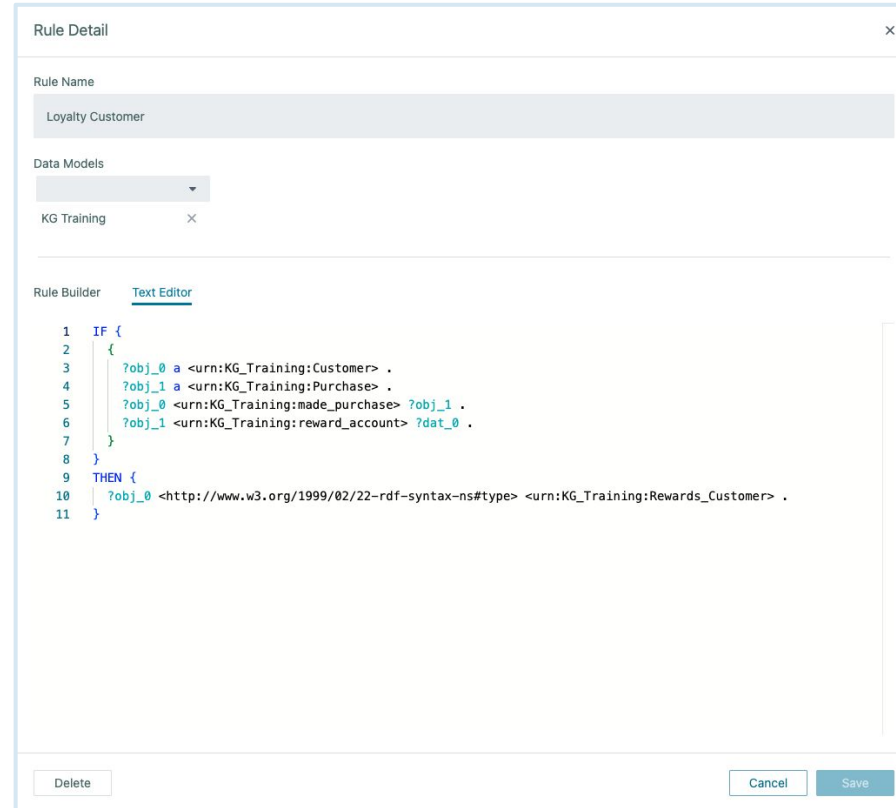
The screenshot shows a 'Rule Detail' window with the following sections:

- Rule Name:** A text input field containing 'New Rule'.
- Data Models:** A dropdown menu showing 'KG Training' with a close button (X).
- Rule Builder / Text Editor:** A tabbed interface with 'Rule Builder' selected.
- When the following conditions are met:** A container for conditions. It currently has one condition: 'Select class...' with a dropdown arrow and a close button (X). Below it is an 'AND NOT' button.
- Then infer:** A container for inference rules. It currently has one rule: 'Select class...' with a dropdown arrow and a close button (X).

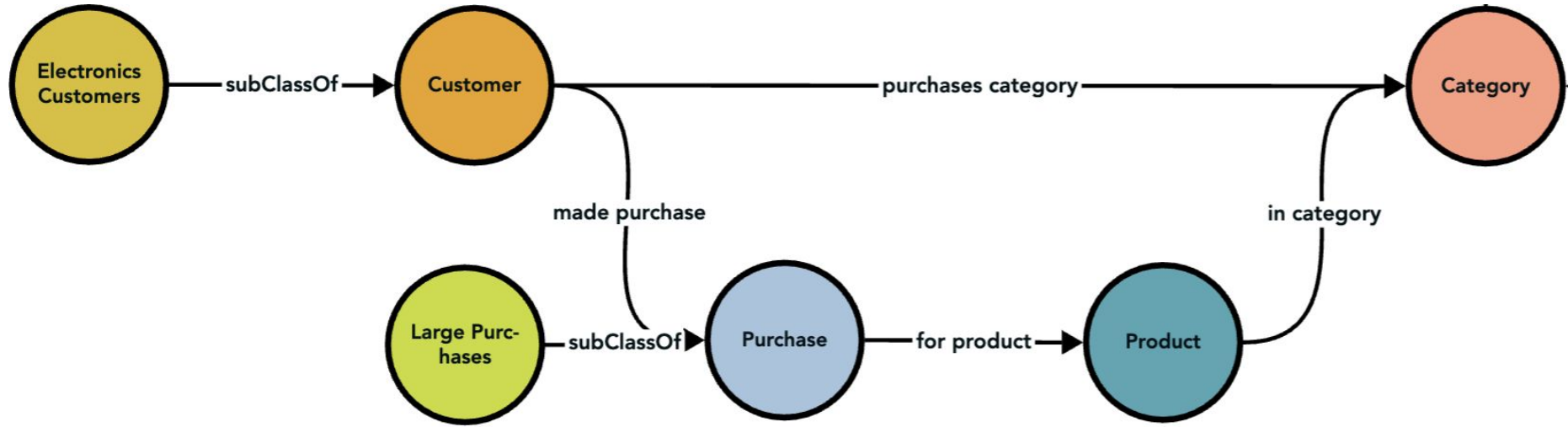
Tip: If you save a change, it will save the changes for the window that was open when saved

Building Rules - Text Editor

- The Text Editor uses the Stardog Rules Syntax (SRS)
- Rules include an IF and Then clause
- Syntax Highlighting
- SRS is based on [W3C SPARQL Grammar](#)

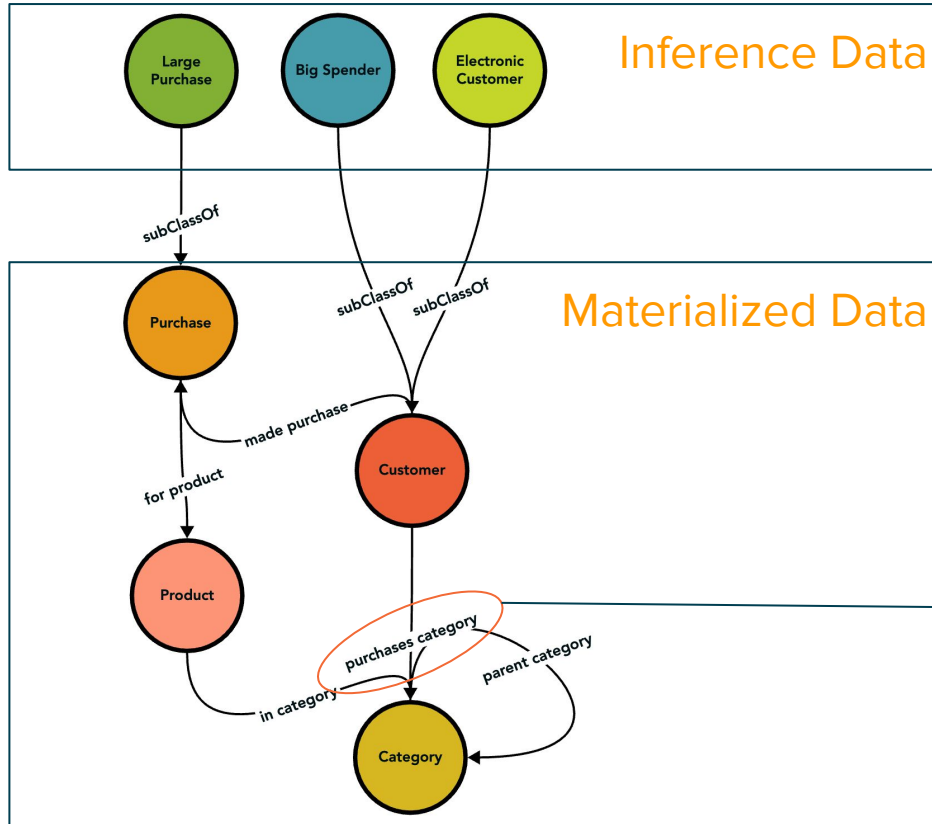


Inference Examples....



- “Electronics Customers” and “Large Purchases” are inferred classes
- “purchases category” is an inferred relationship

Store Data Layers



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Exercise



- Complete Exercise 8 - Create inference rules
- Complete Exercise 9 & 10 - Publish your project and view it in Explorer



Exercise

Instructions

- Open Settings and change the model to **Model you just published** and turn reasoning on
- Build the following query using the query builder. Big Spenders that made a large orders after Jan 1, 2022. Save and run the query



Exercise

- Complete Exercise 11

Introduction to Advanced Designer Options



STARDOG

Terminology

Stardog uses a graph data model based on the Resources Description Framework (RDF), a W3C standard for exchanging graph data.

- **RDF** is a framework for structuring and linking data to describe how things are related.
- **Namespaces** are used as a unique identifier that organizes and categorizes data within a larger set of information.
- **Data Model** is a graph-based structure of triples that represent data relationships. Also referred to as Schema or Ontology.



Namespaces and IRIs

To create unique definitions for data model concepts and instances in an RDF graph, Designer defines IRIs.

Designer allows you to set default values for IRI templates using namespaces and to modify IRIs directly.

IRI

Internationalized Resource Identifier (IRI) is a string used to identify nodes and edges uniquely. Using IRIs prevents name conflicts and doesn't require centralized control.

Class Detail

Class Name

Product

[DATA MODEL NAMESPACE]Product

Relationship Detail

Relationship Name

made purchase

[DATA MODEL NAMESPACE]made_purchase

Example IRI: `tag:stardog:designer:designer_training_example:data:Product123`

Namespace

Local name



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Review Project Settings (Namespace)

Before you begin, review the project settings

- The project name becomes part of the namespace
- If changed, the new namespace will be used for all new model concepts only
- Data Model Settings: sets the namespace
- Choose a project layout

Project Settings

Project Name

DesignerTraining

Layout

☒ Force-directed ☐ Horizontal Tree ☐ Vertical Tree

Customize advanced project details

☒

Instance Data Namespace

urn:DesignerTraining:

Infer class types based on shared properties

☐ Yes ☒ No ☐ Custom per property

Data Model Settings

DesignerTraining Namespace

urn:DesignerTraining:

Review

Course Review

1

Data modeling basics

2

Creating and editing the data model properties

3

Mapping project resources in Designer

4

Publishing the model and viewing in Explorer

5

Inference Rules and Reasoning

Questions?



Thank You!

To contact Stardog about training please email:
training@stardog.com



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